

The 65th ASH Annual Meeting Abstracts

ORAL ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION

Bacterial and Bacteriophage Consortia Are Associated with Protective Intestinal Immuno-Modulatory Metabolites in Allogeneic Stem Cell Transplantation Patients

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The human microbiome is a predictor of clinical outcome in patients undergoing allogeneic hematopoietic stem cell transplantation (allo-SCT). Besides bacteria, fungi and viruses as well as intestinal microbiota-derived metabolites are involved. However, it is still unclear how dynamic shifts in these three kingdoms contribute to the production of intestinal metabolites, how metabolites are impacted by graft-versus-host disease (GvHD) or antibiotics and whether they are associated with clinical outcome.

We report the two-year follow-up of a prospective, observational, longitudinal cohort of allo-SCT patients (n=78) at two German transplantation centers that combined three-kingdom (bacteria, fungi, viruses) analysis of intestinal microbial communities with targeted metabolomics (**Figure 1**). Using Multi-omics factor analysis (MOFA), we uncovered a functional microbiome signature of bacteria from the *Lachnospiraceae* and *Oscillospiraceae* families and their associated bacteriophages, which correlated with the production of immuno-modulatory metabolites (IMMs) including short-chain fatty acids (SCFAs), branched-chain fatty acids (BCFA), metabolites associated with induction of type-I IFN signaling (IIMs) and immuno-modulatory secondary bile acids (**Figure 2**). We established an Immuno-modulatory Metabolite Risk Index (IMM-RI) consisting of five index IMMs, which was associated with improved survival, less transplant-related mortality and reduced relapse rate. Onset of gastrointestinal GvHD and exposure to antibiotics significantly impacted intestinal levels of protective IMMs.

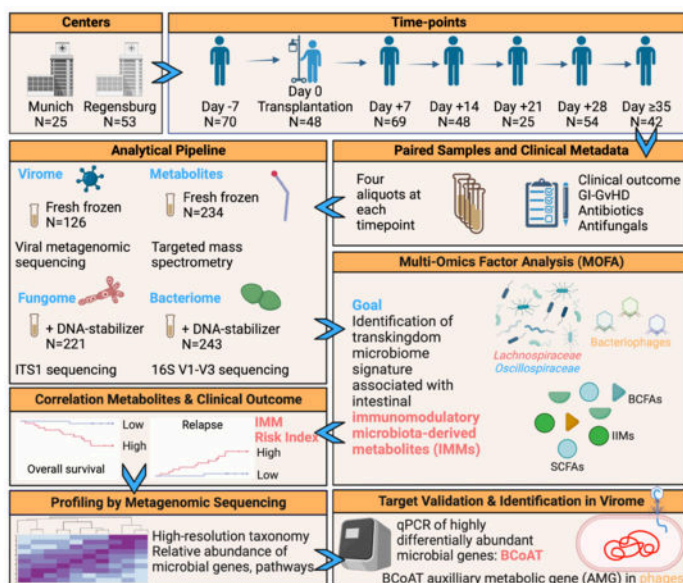
Using whole shotgun metagenomic sequencing, we observed that in IMM-RI low-risk patients, sustained production of protective IMMs was associated with a high abundance of microbial SCFA biosynthesis pathways, specifically butyric acid via

butyryl-CoA:acetate CoA-transferase (BCoAT). Through genome assembly from viral metagenomic sequencing data, we detected two distinct bacteriophages which encoded BCoAT as an auxiliary metabolic gene. They were more abundant in IMM-RI low-risk patients and positively correlated with intestinal butyric acid levels, suggesting that these bacteriophages may modulate bacterial metabolite biosynthesis.

Our study identifies a specific microbiome signature associated with protective IMM that could improve fecal microbiota transplantation (FMT) donor selection and provides a rationale for the development of engineered metabolite-producing consortia and defined metabolite combination drugs as novel microbiome-based therapies for cancer patients.

Disclosures Verbeek: Novartis: Honoraria; Jazz: Honoraria, Other: travel support, Speakers Bureau; MSD: Honoraria, Speakers Bureau; Abbvie: Honoraria, Other: travel support, Speakers Bureau; medac: Other: travel support; Pfizer: Honoraria. **van den Brink:** Nektar Therapeutics: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Pluto Immunotherapeutics: Consultancy, Current holder of stock options in a privately-held company, Honoraria, Membership on an entity's Board of Directors or advisory committees; Frazier Healthcare Partners: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Rheos Medicines: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Seres Therapeutics: Consultancy, Current holder of stock options in a privately-held company, Honoraria, Membership on an entity's Board of Directors or advisory committees, Other: IP licensing, Research Funding; Thymofox: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; DKMS (a non-profit organization): Membership on an entity's Board of Directors or advisory committees; Notch Therapeutics: Consultancy, Current holder of stock options in a privately-held company, Honoraria, Membership on an entity's Board of Directors or advisory committees; Ceramedix: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Lygenesis: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Vor Biopharma: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Da Volterra: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; GlaxoSmithKline: Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; Juno Therapeutics: Other: IP licensing; Wolters Kluwer: Patents & Royalties. **Wolff:** Gilead: Honoraria; Takeda: Honoraria; Behring: Honoraria; Incyte: Honoraria; Mallickrodt: Honoraria; Sanofi: Honoraria; Novartis: Honoraria, Research Funding. **Herr:** Amgen: Honoraria, Other: Travel expenses; Janssen-Cilag: Honoraria, Other: Travel expenses. **Holler:** MaaT Pharma: Consultancy, Membership on an entity's Board of Directors or advisory committees. **Poeck:** Gilead: Consultancy, Honoraria; Abbvie: Consultancy, Honoraria; Pfizer: Consultancy; Novartis: Consultancy, Honoraria; Servier: Consultancy, Honoraria; BMS: Consultancy, Honoraria, Research Funding; Janssen-Cilag: Honoraria.

1



Identification of protective, immune-modulatory metabolites in a two-center, pre-spective, observational, longitudinal cohort of patients undergoing aHS-CT

Figure 1: Study design, sampling scheme and analysis

Illustration of patient numbers per center, sampling time-points, sample numbers per multi-omics analysis, multimodal data analysis, establishment of the IMM-Risk Index as predictor of survival and relapse as well as downstream analysis via metagenomic sequencing and viral genome assembly.

2

Microbiota-derived metabolites

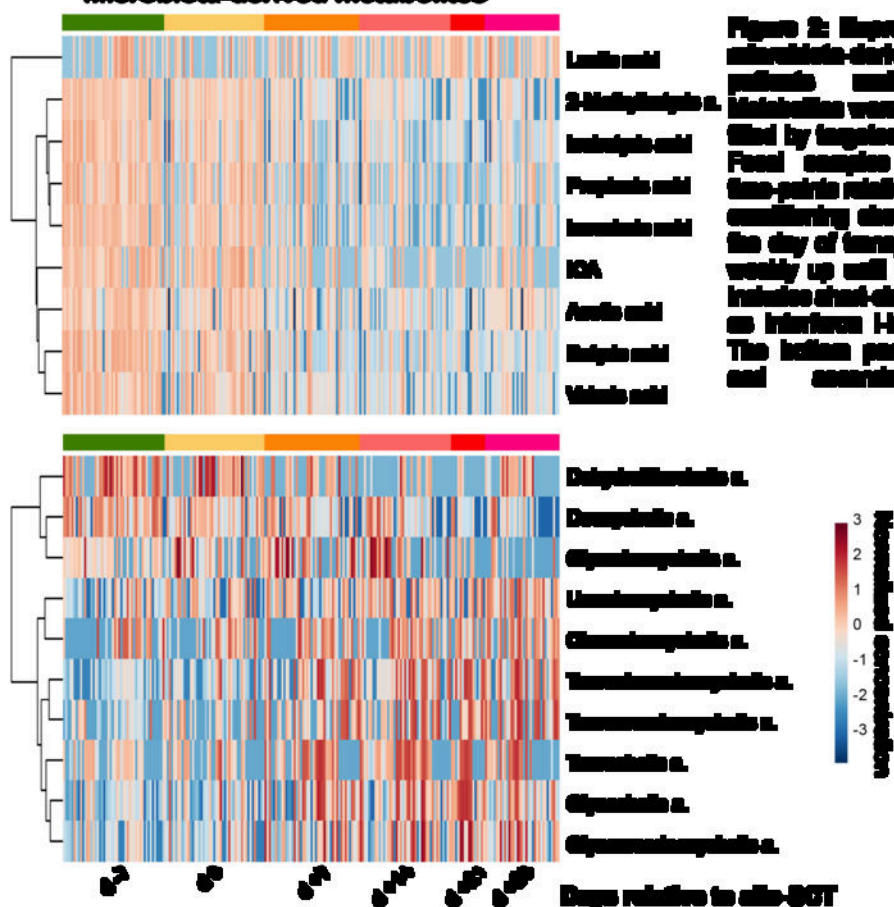


Figure 2: Expression of intestinal microbiota-derived metabolites in patients undergoing aHS-CT
Metabolites were identified and quantified by targeted mass spectrometry. Fecal samples were analyzed at time-points relative to aHS-CT: before considering chemotherapy (d -7), at the day of transplantation (d 0), and weekly up until Day +28. Top panel includes short-chain fatty acids as well as interferon γ -inducing metabolites. The bottom panel contains primary and secondary bile acids.

Figure 1

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